Strict classical repeated-measures ANOVA for the biomarker-treatment interaction: closed-form derivation and limitations

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Purpose. This note derives, in closed form, the strict classical repeated-measures ANOVA (RM-ANOVA) test of the biomarker-treatment interaction for a dichotomized biomarker, states its power formula, and sets out how it relates to the mixed-model Wald test that pmsimstats-ng uses. The derivations are analytic; the few numerical statements are checked against stored simulation output or flagged. The simulation study this note proposed was later carried out as paper 08 (`analysis/report/08-test-procedure-design-sensitivity/`), and dated notes in Sections 6 and 8 point to its results.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. The classical split-plot symbols are *local*; they were renamed on 2026-10-07 so that none collides with a canonical symbol (formerly α_i, β_j, $\pi_{l(i)}$, e_{ijlk}, δ, λ and $1-\beta$).

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
$g \in \{1, \dots, a\}$	biomarker stratum (factor A); $a = 2$ in the main case	local
$j \in \{1, \dots, b\}$	treatment phase (factor B); $b = 2$ in the main case	canonical index
$i \in \{1, \dots, n\}$	participant within stratum	canonical index
$t \in \{1, \dots, T\}$	within-phase timepoint	canonical index
n, N	participants per stratum; total, $N = an$	local; canonical
Y_{gjit}	symptom score (code `Sx`)	canonical
μ, ξ_g, ϕ_j, τ_t	grand mean, stratum, phase and time effects	local
$(\xi\phi)_{gj}$, $(\xi\tau)_{gt}$, $(\phi\tau)_{jt}$, $(\xi\phi\tau)_{gjt}$	interaction effects	local
$u_{i(g)}$, σ_u^2	participant random intercept, nested in stratum, and its variance	canonical
ε_{gjit}, σ_ε^2	within-participant residual and its variance	canonical
$\bar{Y}$ with dots	means over the dotted subscripts	local
SS, MS, df	sums of squares, mean squares, degrees of freedom	local
L, $\hat{L}$	population and sample interaction contrast, eq. (9)	local
L_1	value of L under the alternative	local
Δ_{gi}, $\bar{\Delta}_g$	within-participant phase difference and its stratum mean	local

Symbol	Meaning	Status
F_{AB}, t_{AB}	interaction F- and t-statistics	local
ϵ_{Box}, $\hat{\epsilon}_{\text{Box}}$	Box's sphericity measure and its estimate	local
Λ	non-centrality parameter of the F distribution	local
α, π	significance level; power	canonical
B_i, $D_{bc,it}$, $\beta_{bm:D}$	biomarker, exposure-decayed drug indicator, interaction coefficient	canonical
$t_{1/2}$	carryover half-life	canonical
m	number of levels of the factor whose sphericity is at issue	local

- **Strict RM-ANOVA.** Balanced split-plot analysis of variance with categorical factors, method-of-moments estimation and the sphericity assumption.
- **Sphericity.** All pairwise differences of the within-participant levels have equal variance. Compound symmetry implies it; the converse does not hold. Any factor with two levels is spherical.
- **E1, E2.** The binary on-drug indicator and the exposure-decayed continuous indicator $D_{bc,it}$ (NOTATION.md, Part 3; formerly labeled `A1` and `A2`).
- **Box's measure.** m denotes the number of levels of the within-participant factor whose sphericity is at issue.

1. Motivation

The founding pmsimstats question is the test of the biomarker-treatment interaction in an aggregated N-of-1 trial [15]. The pmsimstats-ng analysis layer fits a continuous-time mixed-effects model [8, 9], in R `nlme::lme(Sx ~ bm + t + Dbc + bm:Dbc)` with a random intercept `~ 1 | ptID` and residual correlation `corCAR1(form = ~ t | ptID)`, and tests the interaction coefficient $\beta_{bm:D}$ via a Wald t-statistic. A reviewer trained in classical analysis-of-variance methodology may reasonably ask whether a strict repeated-measures ANOVA (RM-ANOVA) test of the same interaction is available, what its closed-form expression looks like, and how it relates to the mixed-model test the framework currently produces. This white paper supplies that answer for the strict classical case.

The presentation is deliberately confined to **strict** RM-ANOVA: balanced split-plot design, categorical predictors, sphericity assumption, method-of-moments estimation. Modern extensions (the mixed-effects model for repeated measures, MMRM [10]; generalized estimating equations, GEE [11]; ANCOVA with random participant effects) are mentioned for context but are not strict RM-ANOVA in the textbook sense.

2. Notation and design

Consider the simplest design that supports a biomarker-treatment interaction test in a within-participant setting:

- Two biomarker strata $g \in \{1, 2\}$, formed by dichotomizing the continuous biomarker B_i at a pre-specified cut-point. Strict RM-ANOVA requires categorical predictors; the dichotomization is the price of admission to the framework [12, 13, 14].
- n participants per stratum, balanced, so $N = 2n$ in total. Index $i \in \{1, \dots, n\}$ within stratum.
- Two treatment phases $j \in \{1, 2\}$ (active drug, placebo). In an N-of-1 design these are two phases through which every participant passes; in a parallel-group design they are between-participant and the design is not a split-plot.
- T within-phase timepoints, indexed $t \in \{1, \dots, T\}$. In the simplest case $T = 1$, the analyst pre-averages within-phase observations to a single mean per participant per phase.

The biomarker stratum is **between-participants** (each participant sits in exactly one stratum); the treatment phase and the timepoints are **within-participants** (each participant sees both phases at every timepoint). This is the canonical *split-plot* repeated-measures design [1].

3. The strict RM-ANOVA model

Write Y_{gjit} for the symptom outcome at biomarker stratum g, treatment phase j, participant i within stratum g, and timepoint t. The strict classical split-plot model is

$$\begin{aligned} Y_{gjit} = \mu + \xi_g + \phi_j + (\xi\phi)_{gj} + u_{i(g)} \\ + \tau_t + (\xi\tau)_{gt} + (\phi\tau)_{jt} + (\xi\phi\tau)_{gjt} + \varepsilon_{gjit}, \end{aligned} \tag{1}$$

with sum-to-zero constraints on each fixed effect, the random participant effect $u_{i(g)} \sim N(0, \sigma_u^2)$ nested in stratum, and within-participant error $\varepsilon_{gjit} \sim N(0, \sigma_\varepsilon^2)$, independent. Model (1) therefore implies compound symmetry of

the within-participant covariance: every pair of within-participant observations has the same covariance σ_u^2, and every observation has the same total variance $\sigma_u^2 + \sigma_\varepsilon^2$. The classical F-tests need only the weaker sphericity condition on each within-participant factor (glossary), which compound symmetry implies.

The biomarker-treatment interaction is the $(\xi\phi)_{gj}$ term. The time-resolved version of the interaction is the three-way $(\xi\phi\tau)_{gjt}$ term.

4. Sum-of-squares decomposition

The total sum of squares partitions into between-participants and within-participants components:

$$SS_{\text{tot}} = SS_{\text{between}} + SS_{\text{within}}. \qquad (2)$$

The between-participants component decomposes further into SS_A (biomarker stratum) and $SS_{S(A)}$ (participant within stratum):

$$SS_A = bnT \sum_{g=1}^{a} (\bar{Y}_{g\cdots} - \bar{Y}_{\cdots})^2, \qquad (3)$$

$$SS_{S(A)} = bT \sum_{g=1}^{a} \sum_{i=1}^{n} (\bar{Y}_{g\cdot i\cdot} - \bar{Y}_{g\cdots})^2. \qquad (4)$$

The within-participants component contains SS_B (treatment phase), SS_{AB} (the **biomarker × treatment interaction**, the target of the test), and the within-participants error term for both, $SS_{B\times S(A)}$, together with the time terms of Section 5d:

$$SS_B = anT \sum_{j=1}^{b} (\bar{Y}_{\cdot j\cdot\cdot} - \bar{Y}_{\cdots})^2, \qquad (5)$$

$$SS_{AB} = nT \sum_{g,j} (\bar{Y}_{gj\cdot\cdot} - \bar{Y}_{g\cdots} - \bar{Y}_{\cdot j\cdot\cdot} + \bar{Y}_{\cdots})^2, \qquad (6)$$

$$SS_{B\times S(A)} = T \sum_{g,j,i} (\bar{Y}_{gji\cdot} - \bar{Y}_{g\cdot i\cdot} - \bar{Y}_{gj\cdot\cdot} + \bar{Y}_{g\cdots})^2. \qquad (7)$$

The corresponding degrees of freedom are $df_A = a - 1$, $df_{S(A)} = a(n-1)$, $df_B = b - 1$, $df_{AB} = (a-1)(b-1)$, and $df_{B\times S(A)} = a(n-1)(b-1)$ (the latter being the within-participants error degrees of freedom appropriate for the treatment effect and its interaction with the between-participants factor).

5. The test statistic: closed form

The biomarker-treatment interaction is tested by the F-ratio

$$F_{AB} = \frac{MS_{AB}}{MS_{B\times S(A)}} = \frac{SS_{AB}/df_{AB}}{SS_{B\times S(A)}/df_{B\times S(A)}}. \qquad (8)$$

Under the null $H_0 : (\xi\phi)_{gj} \equiv 0$ and the sphericity assumption on the phase factor, $F_{AB} \sim F(df_{AB}, df_{B\times S(A)})$.

5a. The 2×2 split-plot case

Specialize to $a = b = 2$ (two biomarker strata, two phases). The formulas below keep T general; $T = 1$ is the pre-averaged case. Writing $\bar{Y}_{gj}$ for the cell mean $\bar{Y}_{gj\cdot\cdot}$, the sample interaction contrast is

$$\hat{L} \;=\; (\bar{Y}_{11} - \bar{Y}_{12}) - (\bar{Y}_{21} - \bar{Y}_{22}), \qquad (9)$$

i.e. the difference of within-participant treatment differences across biomarker strata, an estimate of the population contrast L. This is the natural biomarker-treatment interaction statistic.

The interaction parameter under the sum-to-zero ANOVA parameterization satisfies $\widehat{(\xi\phi)}_{11} = \hat{L}/4$ (and the other three cells take values $\pm\hat{L}/4$; derived), so the interaction sum of squares is

$$SS_{AB} \;=\; nT \cdot 4 \cdot (\hat{L}/4)^2 \;=\; \frac{nT\,\hat{L}^2}{4}. \qquad (10)$$

With $df_{AB} = 1$, the interaction mean square equals the interaction sum of squares.

The within-participants error mean square is the pooled estimator of σ_ε^2 from the treatment-by-participant-within-stratum component of variance. Under model (1) and balanced data, $MS_{B\times S(A)}$ is an unbiased estimator $\hat{\sigma}_\varepsilon^2$ of σ_ε^2.

Substituting into equation (8):

$$\boxed{F_{AB} \;=\; \frac{nT\,\hat{L}^2}{4\,\hat{\sigma}_\varepsilon^2}}, \qquad df_1 = 1, \quad df_2 = 2(n-1). \qquad (11)$$

This is the closed-form expression for the strict classical RM-ANOVA test statistic for the biomarker-treatment interaction in a $2 \times 2 \times T$ split-plot design.

5b. Within-participant contrast representation

Equation (11) is equivalent to a two-sample t-test on the within-participant difference. Define $\Delta_{gi} = \bar{Y}_{g1i\cdot} - \bar{Y}_{g2i\cdot}$ as the within-participant treatment difference for participant i in biomarker stratum g, averaged over T within-phase timepoints. Then

$$\hat{L} \;=\; \bar{\Delta}_1 - \bar{\Delta}_2, \qquad \bar{\Delta}_g \;=\; \frac{1}{n}\sum_{i=1}^{n} \Delta_{gi}. \qquad (12)$$

Under model (1), Δ_{gi} are iid within stratum with $\mathrm{Var}(\Delta_{gi}) = 2\sigma_\varepsilon^2/T$ (the random participant effect cancels in the within-participant difference). The variance of $\hat{L}$ is therefore

$$\mathrm{Var}(\hat{L}) \;=\; \frac{2 \cdot 2\sigma_\varepsilon^2}{nT} \;=\; \frac{4\sigma_\varepsilon^2}{nT}. \qquad (13)$$

The pooled-variance two-sample t-statistic for the contrast is

$$t_{AB} \;=\; \frac{\hat{L}}{\sqrt{4\hat{\sigma}_\varepsilon^2/(nT)}} \;=\; \sqrt{\frac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_\varepsilon}, \qquad (14)$$

and $t_{AB}^2 = F_{AB}$, recovering equation (11) (derived). The pooled variance of the Δ_{gi} equals $2\hat{\sigma}_\varepsilon^2/T$, so this is the ordinary pooled t, not a Welch statistic. The strict classical RM-ANOVA F-test for the biomarker-treatment interaction is therefore identical to a two-sample t-test on the within-participant treatment differences, with degrees of freedom $2(n-1)$. Because that t-test needs only that the Δ_{gi} be iid normal with a common variance in the two strata, the identity, and the exactness of the test, hold for any within-participant covariance, AR(1) included, once each phase is reduced to its mean (derived). This identity is well known in the split-plot literature [1, ch. 12; 2] and is the analytical anchor for the statement that the strict-RM-ANOVA test of a biomarker-treatment interaction reduces to a simple two-sample comparison of treatment-effect estimates between biomarker strata.

5c. Generalization: a strata, b phases

For a biomarker strata, b treatment phases, n participants per stratum, and T within-phase timepoints, equation (8) generalizes to

$$F_{AB} = \frac{SS_{AB}/[(a-1)(b-1)]}{SS_{B\times S(A)}/[a(n-1)(b-1)]}. \qquad (15)$$

Under sphericity of the phase factor (automatic when $b = 2$), $F_{AB} \sim F((a-1)(b-1), a(n-1)(b-1))$ under H_0. The closed form of the interaction sum of squares follows directly from equation (6).

5d. The time-resolved three-way case

Including the time factor τ_t explicitly, the test of whether the biomarker modifies the treatment effect across time is the three-way interaction $(\xi\phi\tau)_{gjt}$. The F-statistic is

$$F_{AB\tau} = \frac{MS_{AB\tau}}{MS_{B\tau\times S(A)}}, \qquad (16)$$

with $df_1 = (a-1)(b-1)(T-1)$ and $df_2 = a(n-1)(b-1)(T-1)$. The sums of squares follow the standard split-plot pattern [1, ch. 14].

6. Sphericity, Mauchly, and the Greenhouse-Geisser correction

The F-tests of Section 5 presuppose sphericity of the within-participant factor being tested: all pairwise differences of its levels have equal variance. Compound symmetry is sufficient but not necessary. A factor with two levels is always spherical, so equations (11) and (15) with $b = 2$ are exact whatever the covariance, as shown at the end of Section 5b; sphericity bites on the time-resolved tests of Section 5d, whose within-participant factor has $T > 2$ levels, and on any test with $b > 2$. When sphericity fails, the nominal F-distribution is no longer exact under H_0, and the uncorrected test inflates Type I error. One diagnostic, one summary measure and two corrections are standard:

- **Mauchly's test** [3]. Likelihood-ratio test of sphericity against an unstructured alternative. Often rejects in moderate samples; not informative about the direction of departure.
- **Box's** ϵ_{Box} [4, 5]. A scalar measure of departure from sphericity, with $\epsilon_{\text{Box}} = 1$ under sphericity and $\epsilon_{\text{Box}} \geq 1/(m-1)$ at maximum departure for a factor with m levels ($m = T$ for the time factor).
- **Greenhouse-Geisser correction** [6]. Estimate $\hat{\epsilon}_{\text{Box}}$ from the observed within-participant covariance and multiply both degrees of freedom of the test by it. Conservative.
- **Huynh-Feldt correction** [7]. Less conservative variant, recommended when $\hat{\epsilon}_{\text{Box}} > 0.75$.

The pmsimstats DGP imposes AR(1) within-factor correlation, which is **not** compound-symmetric. Under AR(1), Box's ϵ_{Box} for a factor with more than

two levels is strictly below 1, and the Greenhouse-Geisser correction attenuates the degrees of freedom of the F-tests on that factor. The project audit `docs/06-ar1-residual-correlation.tex` records Type I error of 0.13 (OL) and 0.17 (CO) at $N = 35$ (200 replicates) under the AR(1) DGP for a random-intercept mixed model with conditionally independent residuals (`lme4::lmer`), which imposes the same compound-symmetric working covariance as model (1) (inspected). That figure is indicative only: it is not a simulation of the RM-ANOVA F-test, and the Greenhouse-Geisser corrected Type I error and power were not simulated for this note. The mixed-model analysis with `corCAR1` avoids the issue because it models the AR(1) structure directly.

Correction, 2026-10-07. An earlier version of this section said that equations (11) and (15) presuppose compound symmetry and attributed the 13 to 17 percent figure to strict RM-ANOVA. Paper 08 (`analysis/report/08-test-procedure-design-sensitivity/`, Section 2) makes the same correction and reports that its pre-averaged strict RM-ANOVA arm holds nominal Type I error (inspected).

7. Power calculation under the closed form

For the 2×2 case, equation (11) yields a clean closed-form power formula. Under the alternative $H_1 : L = L_1$, the statistic $F_{AB} = nT\,\hat{L}^2/(4\hat{\sigma}_\varepsilon^2)$ follows a non-central F distribution with non-centrality parameter

$$\Lambda \;=\; \frac{nT\,L_1^2}{4\sigma_\varepsilon^2}. \qquad (17)$$

Power at level α is

$$\pi \;=\; \Pr\Big[F_{AB} > F_{1,2(n-1),1-\alpha} \;\Big|\; \Lambda\Big], \qquad (18)$$

which is computable via standard non-central F functions (`stats::pf` in R with the `ncp` argument; `PROC POWER` in SAS). For sample-size calculation, equation (17) inverts to

$$n \;=\; \frac{4\sigma_\varepsilon^2\Lambda}{TL_1^2}, \qquad (19)$$

with Λ chosen to deliver the target power at the design α. Because the critical value in (18) depends on n through $df_2 = 2(n-1)$, the required Λ itself depends on n, and (19) is solved by iteration (derived).

This formula assumes (i) the variance $2\sigma_\varepsilon^2/T$ of each phase difference implied by model (1), (ii) categorical biomarker at the dichotomized cut-point with equal stratum sizes, (iii) no carryover, (iv) no within-treatment relapse, (v) iid within-phase observations after within-participant differencing. All five assumptions

are violated to varying degrees in the pmsimstats DGP; see Section 8. Under AR(1) errors the test stays exact (Section 5b), but the variance of Δ_{gi} is no longer $2\sigma_\varepsilon^2/T$, so (17) must use the actual variance of the phase difference.

8. Limitations of the strict classical formulation

The closed-form simplicity of equation (11) is bought with five strong assumptions, each of which is empirically problematic in the pmsimstats setting.

1. **Categorical biomarker.** The pmsimstats biomarker (resting SBP in the prazosin-PTSD application) is continuous and has no defensible cut-point. Dichotomizing at the median or at a clinical threshold loses the within-stratum biomarker variation, attenuates the interaction estimator, and inflates Type II error [12, 13, 14]. The information loss can be substantial. An earlier version of this point quoted a median split standard error 1.4 to 1.7 times the continuous one and a 30 to 50 percent power penalty "in the carryover-sensitivity production data"; no script or stored output producing those figures was found (2026-10-07), so they are withdrawn. Paper 08 has since measured the cost directly: replacing the dichotomized biomarker by the continuous one raises power by 0.08 to 0.21 across its sample sizes (0.180 at $N = 25$), while the further step to the full mixed model adds at most 0.04 (paper 08 abstract, inspected).

2. **Sphericity.** Compound-symmetric within-participant covariance is empirically rejected for time-series symptom data: nearby timepoints are more correlated than distant ones. The AR(1) structure that pmsimstats implements violates sphericity for any within-participant factor with more than two levels. The pre-averaged two-phase test of equation (11) is unaffected (Section 6); the time-resolved test of Section 5d needs a Greenhouse-Geisser or Huynh-Feldt correction, which restores approximately nominal Type I error at a loss of degrees of freedom and power.

3. **No carryover model.** Strict RM-ANOVA has no provision for residual drug effect during washout; the model treats off-drug observations as if no drug had been administered. This is the binary-indicator specification E1 of the carryover-sensitivity manuscript (paper 02). In its stored grid (`analysis/data/02-grid-summary.rds`, $N = 70$, 500 replicates per cell, exponential decay), E1 at $t_{1/2} = 1.0$ week attenuates $\hat{\beta}_{bm:D}$ toward zero by 30 to 34 percent in the Hybrid design and by 40 to 42 percent in OL+BDC under covariance moderation, but by at most 4 percent in CO and by at most 1.4 percent in any design under mean moderation (verified, 2026-10-07). The attenuation of $\hat{L}$ is expected to follow the same pattern (argued, not simulated).

4. **Balanced design required.** Equations (11) and (15) assume equal stratum sizes and equal numbers of within-phase timepoints. Real trials have

missing data; classical RM-ANOVA handles missingness only through complete-case deletion. The MMRM framework's likelihood-based handling of MAR data [10] is better behaved.

5. **Equal time spacing.** Strict RM-ANOVA treats time as a categorical factor with implicitly equal-spaced levels. Real N-of-1 trials have unequally-spaced visits; pmsimstats implements continuous-time AR(1) (`corCAR1`), which the strict-RM-ANOVA framework cannot do.

9. Relationship to the pmsimstats MMRM analysis

The strict-RM-ANOVA F-statistic in equation (11) and the pmsimstats `nlme::lme` Wald t-statistic on `bm:Dbc` are **asymptotically equivalent** under three conditions: (i) the biomarker enters the mixed model as the stratum indicator, (ii) the within-participant covariance is exactly compound-symmetric, and (iii) the drug indicator is binary (specification E1 rather than the exposure-decayed $D_{bc,it}$ of E2). When all three hold, the two tests reach the same coefficient and the same standard error up to small-sample correction factors (argued). When any fails, the mixed-model test extracts more information than the strict RM-ANOVA: the continuous biomarker contributes within-stratum variation, the AR(1) covariance is modeled directly, and the continuous $D_{bc,it}$ exposure-decay predictor (E2) uses the off-drug trajectory.

For the founding pmsimstats interaction question [15], the strict-RM-ANOVA test is therefore a **lower-power reduction** of the available analysis rather than a competing alternative. Where a reviewer requests 'RM-ANOVA results', the natural response is one of (a) point at the existing mixed-model analysis as a continuous-time member of the repeated-measures family, (b) report the strict-RM-ANOVA test from equation (11) as a sensitivity comparator under dichotomized biomarker and binary phase, or (c) report both with the explicit power-loss tabulation.

10. Recommendation for pmsimstats

We recommend retaining the MMRM-with-continuous-time analysis as primary and adding a strict-RM-ANOVA comparator as a planned sensitivity:

1. **Categorize the biomarker at the median.** Conventional choice; the alternative pre-specified clinical cut-point is acceptable when one exists.
2. **Pre-average within-phase timepoints to a single mean per participant per phase.** Reduces the T-axis to $T = 1$ for the sensitivity test and makes the test exact under any within-participant covariance (Section 5b).

3. **Compute $\hat{L}$ per equation (9) and F_{AB} per equation (11).** This is computable by hand from the four cell means and the within-participant error mean square.
4. **Report the strict-RM-ANOVA p-value alongside the mixed-model p-value in the manuscript table.** The two should be in the same order of magnitude. Substantial divergence between them indicates that the dichotomization, the pre-averaging, or the binary phase indicator is doing meaningful work, and the discrepancy is informative for interpretation.

The carryover-sensitivity manuscript at `analysis/report/02-carryover-sensitivity/` already evaluates the E1 binary-phase specification, which is the analysis-model counterpart to the strict-RM-ANOVA closed form derived here. The manuscript reports the E1 specification's bias and coverage under carryover. The strict-RM-ANOVA test is expected to inherit the E1 specification's attenuation properties (argued; see Section 8, point 3). Paper 08 implements this comparator as its `rmanova` arm (`analysis/scripts/test-procedure-design-sensitivity/01-study1-test-procedure.R`).

References

Metadata were checked on 2026-10-07 against PubMed (entries 8, 12 to 15) and Crossref (entries 4 to 7, 10); entries 1, 2 and 9 are books and entries 3 and 11 are cited from their standard records. An uncited Gompertz (1825) entry was removed.

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